

Unifying Special Relativity and Quantum Mechanics

Lecture 23 — Semi-Simple Gauge Groups, the Matter Sector, and the Failure of Massive Yang–Mills

(Block-diagonal adjoints, one coupling per factor, tensor-product representations, Fermi theory, and why a gauge-boson mass cannot simply be written down)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–22: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, pencil). Everything inside a blue box like this one is material I have added; it is *not* in the original notes.

Pagination. For the first time in a while there is *no* restart: this scan is the direct continuation of Lecture 22, picking up at [p. 25*****] exactly where that one broke off at [p. 24*****], in the middle of the discussion of semi-simple groups. The eleven-star numbering therefore carries straight on: [p. 25*****]–[p. 36*****]. The scan is in order, complete, and free of duplicates.

Continuity with Lecture 22. The closing blue box of the Lecture 22 document anticipated the result that this lecture derives properly — one independent coupling per simple factor. Sheets [p. 25*****]–[p. 28*****] are the actual argument, and it is more interesting than the bare statement: the reason is that the adjoint representation becomes *reducible*, so the gauge fields split into blocks that never mix.

Two conventions again. As in Lecture 22, the coupling constants move when the fields are rescaled at [p. 32*****]. Before: $-\frac{1}{4g_i^2}F_iF_i$ with no g in D_μ . After $W \rightarrow g_1W$, $G \rightarrow g_2G$, $A \rightarrow gA$: $-\frac{1}{4}F_iF_i$ with the couplings inside D_μ . Same theory.

Where the scan stops. Sheet [p. 36*****] ends by asking what new states must appear at the cutoff to complete a massive Yang–Mills theory. The question is left open — deliberately, since the answer is the Higgs mechanism. A closing blue box says what is coming without pre-empting it.

Where we left off — recap of Lecture 22

Lecture 22 built non-abelian gauge theory in three steps. Starting from a global compact symmetry $\Phi \rightarrow U\Phi$ with $[T^A, T^B] = if_{ABC}T^C$, promoting $\theta_A \rightarrow \theta_A(x)$ broke the derivative term; repairing it forced $\dim G$ gauge fields into a covariant derivative $D_\mu\Phi = \partial_\mu\Phi + iG_\mu^AT^A\Phi$ with the transformation law

$$G_\mu^AT^A \longrightarrow i(\partial_\mu U)U^{-1} + U(G_\mu^AT^A)U^{-1}.$$

Infinitesimally, the second term says the gauge fields transform in the *adjoint representation*, with generators $(t^B)_{AC} = -if_{BAC}$. The field strength, defined through $[D_\mu, D_\nu]\Phi = iF_{\mu\nu}^AT^A\Phi$, came out as $F_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC}G_\mu^BG_\nu^C$ and transforms in the adjoint too, so $\sum_A F_{\mu\nu}^AF^{A,\mu\nu}$ is invariant. Expanding it gave photon-like kinetic terms plus cubic and quartic self-interactions, all controlled by a single dimensionless g .

This lecture does three things with that machinery. First (pp. 25–29) it completes the semi-simple case, showing *why* each simple factor gets its own coupling. Second (pp. 30–32) it sets

up the matter sector when several groups act at once. Third (pp. 32–36) it asks the question that the whole construction has been building towards: can the gauge bosons have a mass? The answer, on the last sheet, is no — not by writing one down — and the failure is the same unitarity failure that Lecture 21 diagnosed for gravity.

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PART I — Semi-simple groups: why one coupling per factor (pp. 25*****–29*****)

1 The adjoint: irreducible or reducible [p. 25*****]

[p. 25*****]

The key point is precisely the structure of the adjoint representation:

- i) If the group G is simple, the adjoint representation is irreducible
- ii) If the group G is semi-simple, the adjoint representation is reducible

$\hookrightarrow$ and it is given by the direct sum of the adjoint representations that correspond to each of the simple subgroup

To visualize this point, let us consider $F_{\mu\nu}^A$.

Under G , it transforms according to

$$F_{\mu\nu}^A \xrightarrow{G} F_{\mu\nu}'^A = F_{\mu\nu}^A - i\theta_B (t^B)_{AC} F_{\mu\nu}^C$$

This is the infinitesimal transformation we found before.

If the transformation is finite, the above correction as usual exponentiates:

$$F_{\mu\nu}^A \xrightarrow{G} F_{\mu\nu}'^A = \left[\exp(-i\theta_B t^B) \right]_{AC} F_{\mu\nu}^C$$

$\uparrow$ generators in the adjoint of G

What “reducible” means, and why it is the whole point

A representation is *reducible* if the vector space it acts on splits into subspaces that the group never mixes. In a suitable basis the representation matrices are then block diagonal, and each block is a representation in its own right.

The claim here is a statement about the structure constants. For $G = G_1 \times \cdots \times G_m$, generators belonging to different factors *commute*:

$$[T^{A_i}, T^{B_j}] = 0 \quad \text{for } i \neq j \quad \implies \quad f_{ABC} = 0 \text{ unless } A, B, C \text{ all lie in the same factor.}$$

Since the adjoint generators *are* the structure constants, $(t^B)_{AC} = -if_{BAC}$, the adjoint matrices inherit exactly this block structure. A gauge field belonging to G_2 is simply never rotated into one belonging to G_3 .

Why this matters so much. In Lecture 22 the single coupling g was forced because $\sum_A F^A F^A$ was a *single* invariant — there was no way to weight one generator differently from another without breaking the symmetry. When the adjoint is reducible that argument weakens: each block gives a separate invariant, so each may carry its own coefficient. Hence m couplings. The next three sheets make this visual.

2 Block-diagonal structure [p. 26*****–p. 27*****]

[p. 26*****]

The fact that the adjoint representation is reducible means that the above matrix transformation is block diagonal

Schematically

the full matrix is $\dim G \times \dim G$ dimensional

Each non-vanishing block is $\dim G_i \times \dim G_i$ -dimensional, and the transformation matrix takes the form

$$\exp(-i\theta_A t_i^A)$$

where t_i^A are the generators in the adjoint representation of G_i

Each block transforms independently from the others. We then group together objects that do not mix under the transformation:

[p. 27*****]

The important consequence is that the quantity

$$\sum_{A=1}^{\dim G_i} F_{i,\mu\nu}^A F_i^{A,\mu\nu}$$

for each $i = 1, \dots, m$

is invariant under the action of the full group G . Consequently, the pure Yang–Mills Lagrangian is

Reading the two diagrams

The first picture is the general statement: the $\dim G \times \dim G$ transformation matrix has non-zero entries only in blocks along the diagonal. The second is the same picture after relabelling, with the components of $F_{\mu\nu}^A$ sorted so that all fields belonging to G_1 come first, then those of G_2 , and so on. The i -th block is the adjoint representation of G_i alone, of size $\dim G_i \times \dim G_i$, and it acts only on the fields of that factor.

Everything else follows. Because block i never talks to block j , the length of each sub-vector is separately preserved: $\sum_A F_i^A F_i^A$ is invariant on its own. There are m independent invariants where a simple group would have had one.

Note the index gymnastics in the notation: $F_{i,\mu\nu}^A$ carries three labels — i says which simple factor, A runs over the generators of that factor (1 to $\dim G_i$), and $\mu\nu$ are the Lorentz indices. Only A is summed in the invariant.

3 The pure Yang–Mills Lagrangian, and $SU(2) \times SU(3)$ [p. 28***]**

[p. 28*****]

$$\mathcal{L}_{YM} = \sum_{i=1}^m -\frac{1}{4g_i^2} \sum_{A=1}^{\dim G_i} F_{i,\mu\nu}^A F_i^{A,\mu\nu}$$

↔ The key aspect is that, since each term is individually invariant, we are forced to introduce m free parameters $g_1, \dots, g_m$

Example

We consider the case $G = SU(2) \times SU(3)$

We have

$$\left. \begin{array}{l} \dim SU(2) = 3 \\ \dim SU(3) = 8 \end{array} \right\} \quad \blacktriangleright \quad \dim G = 11$$

We introduce 3 gauge bosons for $SU(2)$: $W_\mu^{A=1,2,3}$

8 gauge bosons for $SU(3)$: $G_\mu^{A=1,\dots,8}$

We have the field strength tensors

$$W_{\mu\nu}^{A=1,2,3} = \partial_\mu W_\nu^A - \partial_\nu W_\mu^A - \varepsilon_{ABC} W_\mu^B W_\nu^C$$

ε_{ABC} structure constant of $SU(2)$

$$G_{\mu\nu}^{A=1,\dots,8} = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC} G_\mu^B G_\nu^C$$

f_{ABC} structure constants of $SU(3)$

The pure Yang–Mills Lagrangian is

[p. 29*****]

$$\mathcal{L}_{YM} = -\frac{1}{4g_2^2} W_{\mu\nu}^A W^{A,\mu\nu} - \frac{1}{4g_3^2} G_{\mu\nu}^A G^{A,\mu\nu}$$

Why the couplings cannot be related, and what that costs

Nothing in gauge invariance relates g_2 to g_3 . Each term is separately invariant, so each separately admits an arbitrary coefficient, and the theory simply does not predict their ratio. They must be measured.

This is not a small point. In the Standard Model it means three independent numbers g_3, g_2, g_1 — roughly 1.2, 0.65, 0.36 at the weak scale — that the theory takes as input. The observation that, when run to very high energy, they *nearly* meet at a single value around 10^{16} GeV is the main quantitative motivation for grand unification: if the three factors were embedded in one *simple* group (like $SU(5)$ or $SO(10)$), the argument of [p. 25*****] would apply, the adjoint would be irreducible, and a *single* coupling would be forced.

A note on the subscripts. The notes label the couplings g_2 and g_3 after the group ($SU(2)$, $SU(3)$), not after the index i of the sum. Both conventions appear in the literature; here the $SU(2)$ field strength is called W and the $SU(3)$ one G , which is deliberate — these are precisely the names of the weak and strong gauge fields in the Standard Model.

4 Additional $U(1)$ factors [p. 29*****]

► Semi-simple Lie groups with additional $U(1)$ factors

The most general situation is the one in which

$$G = \underbrace{G_1 \times \cdots \times G_m}_{\text{simple factors}} \times \underbrace{U(1) \times \cdots \times U(1)}_{\text{additional } U(1) \text{ factors}}$$

↓ simple factors

↓ additional $U(1)$ factors

The construction generalizes to this case: for each $U(1)$ factor, we simply add a gauge boson and the corresponding field strength (which is the standard abelian one).

↪ The pure Yang–Mills Lagrangian reads

$$\mathcal{L}_{YM} = \sum_{i=1}^m \frac{-1}{4g_i^2} \sum_{A=1}^{\dim G_i} F_{i,\mu\nu}^A F_i^{A,\mu\nu} - \frac{1}{4g^2} F_{\mu\nu} F^{\mu\nu}$$

$$\longrightarrow F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

→ one term of this kind for each abelian $U(1)$ factor

Why $U(1)$ has to be listed separately

Strictly, $U(1)$ is *not* simple — it is abelian, and an abelian group is excluded from the classification of Lecture 22 [p. 2*****], which covered non-abelian simple groups. So a product like $SU(3) \times SU(2) \times U(1)$ is not semi-simple in the mathematician's sense, and the notes are right to give the $U(1)$ factors their own line.

Physically the abelian factor is the easy case: $f_{ABC} = 0$, so $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ with no quadratic term, no self-interaction, and the gauge boson is neutral. It is just electromagnetism bolted on.

One genuine difference is worth knowing, even though it is beyond this lecture: for a simple factor, gauge invariance quantizes the charges — the representations of $SU(N)$ come in a

discrete list, so the possible “charges” are fixed. For a $U(1)$ factor, the charge q appearing on the next sheet is a free real number, and nothing forbids any value. That the observed hypercharges of the Standard Model come out in neat rational ratios (which is what makes the proton and electron charges cancel to one part in 10^{21}) is therefore unexplained by gauge invariance alone — another argument pointing to unification in a simple group.

PART II — The matter sector (pp. 30*****–32*****)

5 Specifying the representations [p. 30*****]

[p. 30*****]

► The matter sector

The structure is

$$\mathcal{L} = \mathcal{L}_{YM} + \mathcal{L}_M(\Phi, D_\mu \Phi)$$

If $G = G_1 \times \cdots \times G_m \times U(1)$ we need to specify how Φ transform according to each group G_i and $U(1)$ factor

Let us discuss this situation with the help of a concrete example.

↔ Suppose that $G = G_1 \times G_2 \times U(1)$ and suppose that we have a set of fields that transform:

- i) according to the representation “ R_1 ” of G_1 with generators $T_{R_1}^A$ ($A = 1, \dots, \dim G_1$)
- ii) according to the representation “ R_2 ” of G_2 with generators $T_{R_2}^A$ ($A = 1, \dots, \dim G_2$)
- iii) according to the representation of $U(1)$ with charge q

The field carries two indices, and we write the transformations as follows:

[p. 31*****]

$$\Phi^{a\alpha} \xrightarrow{G_1} \left(\exp(-i\theta_A T_{R_1}^A) \right)^a_b \Phi^{b\alpha}$$

- $A = 1, \dots, \dim G_1$
- $a, b = 1, \dots, \dim R_1$

$$\Phi^{a\alpha} \xrightarrow{G_2} \left(\exp(-i\tilde{\theta}_A T_{R_2}^A) \right)^\alpha_\beta \Phi^{a\beta}$$

- $A = 1, \dots, \dim G_2$
- $\alpha, \beta = 1, \dots, \dim R_2$

$$\Phi^{a\alpha} \xrightarrow{U(1)} \exp(-iq\theta) \Phi^{a\alpha}$$

► In matrix form, Φ is a vector with dimension $\dim R_1 \times \dim R_2$ and the transformations formally act on it as

$$\begin{aligned}\Phi &\xrightarrow{G_1} \left(\exp(-i\theta_A T_{R_1}^A) \otimes \mathbf{1}_{R_2} \right) \Phi \\ \Phi &\xrightarrow{G_2} \left(\mathbf{1}_{R_1} \otimes \exp(-i\tilde{\theta}_A T_{R_2}^A) \right) \Phi \\ \Phi &\xrightarrow{U(1)} \exp(-iq\theta) \mathbf{1}_{R_1 \times R_2} \Phi\end{aligned}$$

The Lagrangian is given by

$$\mathcal{L}_{YM} = -\frac{1}{4g_1^2} W_{\mu\nu}^A W^{A,\mu\nu} - \frac{1}{4g_2^2} G_{\mu\nu}^A G^{A,\mu\nu} - \frac{1}{4g^2} F_{\mu\nu} F^{\mu\nu} + \mathcal{L}_M(\Phi, D_\mu \Phi)$$

pure Y-M terms

The tensor product, made concrete

The abstract statement is that a field charged under several groups at once lives in a *tensor product* of representations, and carries one index per factor. If Φ is in R_1 of G_1 (dimension $\dim R_1$) and R_2 of G_2 (dimension $\dim R_2$), the total number of components is the *product* $\dim R_1 \times \dim R_2$.

The $\otimes \mathbf{1}$ structure says that a G_1 transformation acts on the first index and leaves the second alone. Two consequences follow immediately: the transformations of different factors commute, and each factor's generators act “blindly” with respect to the others.

The Standard Model example. A left-handed quark doublet is

$$Q_L = (\mathbf{3}, \mathbf{2})_{1/6},$$

i.e. the fundamental (3) of colour $SU(3)$, the fundamental (2) of weak $SU(2)$, and hypercharge $q = 1/6$. It therefore has $3 \times 2 = 6$ components (each still a two-component Weyl spinor): up-red, up-green, up-blue, down-red, down-green, down-blue. A gluon rotates the colour index and leaves the weak index untouched; a W rotates up $\leftrightarrow$ down and leaves colour untouched. That is exactly the $\otimes \mathbf{1}$ structure written above, and it is why the notation is worth the trouble.

6 The covariant derivative and the rescaling [p. 32*****]

[p. 32*****]

↓ For the covariant derivative, we write

$$D_\mu \Phi = \partial_\mu \Phi + iW_\mu^A T_{R_1}^A \Phi + iG_\mu^A T_{R_2}^A \Phi + iq A_\mu \Phi$$

↖ short hand notation for $T_{R_1}^A \otimes \mathbf{1}_{R_2}$

↖ short hand for $\mathbf{1}_{R_1} \otimes T_{R_2}^A$

If we rescale the fields, $W_\mu^A \rightarrow g_1 W_\mu^A$; $G_\mu^A \rightarrow g_2 G_\mu^A$; $A_\mu \rightarrow g A_\mu$

$$\mathcal{L} = -\frac{1}{4}W_{\mu\nu}^A W^{A,\mu\nu} - \frac{1}{4}G_{\mu\nu}^A G^{A,\mu\nu} - \frac{1}{4}F_{\mu\nu} F^{\mu\nu} + \mathcal{L}_M(\Phi, D_\mu \Phi)$$

$$D_\mu \Phi = \partial_\mu \Phi + ig_1 W_\mu^A T_{R_1}^A \Phi + ig_2 W_\mu^A T_{R_2}^A \Phi + ig q A_\mu \Phi$$

One derivative, three couplings

The covariant derivative simply accumulates one term per factor of the group. Each carries its own coupling and its own generators, and each acts on its own index. This single expression is the complete statement of how matter couples to gauge fields in the Standard Model:

$$D_\mu = \partial_\mu + ig_3 G_\mu^A T_{\text{colour}}^A + ig_2 W_\mu^A T_{\text{weak}}^A + ig_1 Y B_\mu.$$

Everything about which particle feels which force is encoded in the choice of representations: a right-handed electron is a colour singlet ($T_{\text{colour}}^A = 0$, so it does not feel the strong force) and a weak singlet, but has non-zero hypercharge, so only the third term survives for it.

Note the $U(1)$ asymmetry. For the non-abelian factors the coupling and the generator are separate objects; for the abelian one they merge into the product gq . Only that product is measurable, which is why the normalization of hypercharge is conventional — one can rescale $q \rightarrow \lambda q$, $g \rightarrow g/\lambda$ with no physical consequence. This is another way of seeing the charge-quantization point from [p. 29*****].

PART III — Is there a consistent theory of massive Yang–Mills? (pp. 32*****–36*****)

7 Fermi’s theory of neutron decay [p. 32*****–p. 33*****]

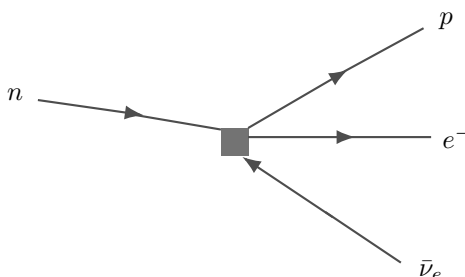
■ Is there a consistent theory of massive pure Yang–Mills?

To explain the neutron decay, Fermi formulated the theory based on the Lagrangian interaction given by:

[p. 33*****]

$$\mathcal{L}_{int} = -\frac{G_F}{\sqrt{2}} \underbrace{(\bar{p} \gamma_\mu n)(\bar{e} \gamma^\mu \nu)} + \text{h.c.}$$

↓ “ n ” destroy the neutron, “ $\bar{p}$ ” create a proton, “ $\bar{e}$ ” create the electron, “ ν ” create the antineutrino



↓ This is a non-ren. interaction ($[G_F] = [M]^{-2}$) and we expect that it is generated by the exchange of an heavy particle in the limit in which the energy of the process is well below its mass.

↪ Two important observations

i) Because of the γ -structure of the interaction, we expect the exchanged particle to be spin-1 (and not spin-0)

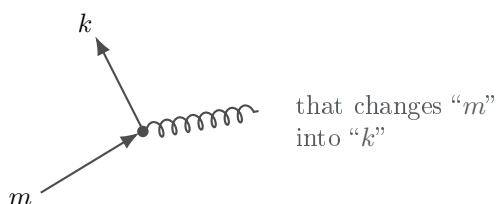
Schematically

$$(\bar{p}\gamma^\mu n) \frac{i}{p^2 - M^2 + i\epsilon} \left(-g_{\mu\nu} + \frac{p_\mu p_\nu}{M^2} \right) (\bar{e}\gamma^\nu \nu) \longrightarrow (\bar{p}\gamma^\mu n) \frac{i g_{\mu\nu}}{M^2} (\bar{e}\gamma^\nu \nu)$$

[p. 34*****]

ii) The interaction must change the nature of the interacting particles

This is precisely what a non-abelian gauge theory does, remember the vertex



These observations point towards the investigation of massive Yang–Mills theories.

Why these two observations are exactly the right clues

Fermi's interaction is a four-fermion contact operator — precisely the shape of the toy model of Lecture 21 [p. 21*****], with G_F playing the role of c . That lecture established the moral: a non-renormalizable contact interaction is the low-energy shadow of something heavier. The two observations here identify *what*:

(i) **Spin 1, from the index structure.** The Fermi operator is a product of two *vector* currents $\bar{\psi}\gamma^\mu\psi$, with a Lorentz index on each. A spin-0 exchange would produce $(\bar{\psi}\psi)(\bar{\psi}\psi)$ with no index — the wrong structure. Only a spin-1 propagator, whose numerator $-g_{\mu\nu} + p_\mu p_\nu/M^2$ carries two indices, can tie the two currents together this way. And in the low-energy limit $p^2 \ll M^2$ the propagator collapses to the constant $ig_{\mu\nu}/M^2$, reproducing the contact interaction with the matching $G_F/\sqrt{2} \sim g^2/M^2$.

(ii) **It must change particle type.** The vertex turns a neutron into a proton, and an electron into a neutrino. A neutral spin-1 boson like the photon cannot do this. But the vertex $-ig\gamma^\mu(T^A)_{km}$ derived at the very end of Lecture 22 does exactly this, because T^A is off-diagonal.

Put together: the mediator must be a *massive, non-abelian gauge boson*. That is the W . So the entire logical pressure of the argument is towards a Yang–Mills theory whose gauge bosons have a mass — and the rest of this lecture asks whether that is even possible.

The $p_\mu p_\nu/M^2$ term is worth watching. It is harmless here because $p_\mu \bar{p}\gamma^\mu n$ is suppressed by the light fermion masses — but notice it is singular as $M \rightarrow 0$, and it is exactly this term that will blow up on [p. 36*****].

8 Naive massive Yang–Mills [p. 34*****–p. 35*****]

We try a naive approach.

We consider the theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^A F^{A,\mu\nu} + \frac{1}{2}M^2 G_\mu^A G^{A,\mu}$$

↗ pure Yang–Mills with $F_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A - f_{ABC}G_\mu^B G_\nu^C$

↖ like a Proca mass term for each of the gauge fields

This theory is not invariant under gauge transformations

$$G_\mu^A T^A \longrightarrow i(\partial_\mu U)U^{-1} + U(G_\mu^A T^A)U^{-1}$$

[p. 35*****]

↔ Not a problem, since the gauge bosons are massive we do not need gauge invariance

The theory retains invariance under global transformations, under which both $F_{\mu\nu}^A$ and G_μ^A transform according to the adjoint.

Is this theory consistent?

Why a mass term breaks gauge invariance

Under a gauge transformation the mass term picks up the inhomogeneous piece: schematically $G_\mu \rightarrow U G_\mu U^{-1} + i(\partial_\mu U)U^{-1}$, so

$$G_\mu^A G^{A,\mu} \longrightarrow G_\mu^A G^{A,\mu} + (\text{terms involving } \partial_\mu U) \neq G_\mu^A G^{A,\mu}.$$

The homogeneous part alone would have been fine — a trace of two adjoint objects is invariant — but the $\partial_\mu U$ term spoils it. The same is true in QED: a photon mass $\frac{1}{2}M^2 A_\mu A^\mu$ is not invariant under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$. Gauge invariance forbids gauge-boson masses, full stop.

The green remark “not a problem, we do not need gauge invariance” deserves scrutiny, because it is the step the rest of the lecture demolishes. It is not unreasonable on its face: gauge invariance is a *redundancy* of description, not a symmetry with physical content, and a massive vector field is perfectly well defined on its own (Proca theory — three physical polarizations instead of two). For an *abelian* theory this really is fine: massive QED is consistent.

What goes wrong is specific to the non-abelian case, and it is the extra longitudinal polarization interacting with the self-interactions that gauge invariance used to organize. That is the calculation on the next sheet.

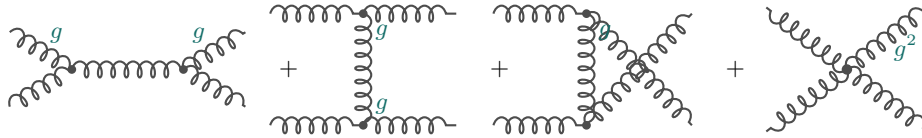
9 Longitudinal scattering and the loss of unitarity [p. 35*****—p. 36*****]

There is a problem. Since G_μ^A are now massive, we open up the possibility of studying scattering of longitudinally polarized gauge bosons

Consider the process

$$G_L^a(p_1) + G_L^b(p_2) \longrightarrow G_L^c(p_3) + G_L^d(p_4)$$

In full generality, we have the diagrams



The amplitude has the structure

$$i\mathcal{M} = \varepsilon_L^\mu(p_1) \varepsilon_L^\nu(p_2) \varepsilon_L^\rho(p_3)^* \varepsilon_L^\sigma(p_4)^* i\mathcal{M}_{\mu\nu\rho\sigma}$$

We consider the ultra-relativistic limit $E \gg M$

[p. 36*****]

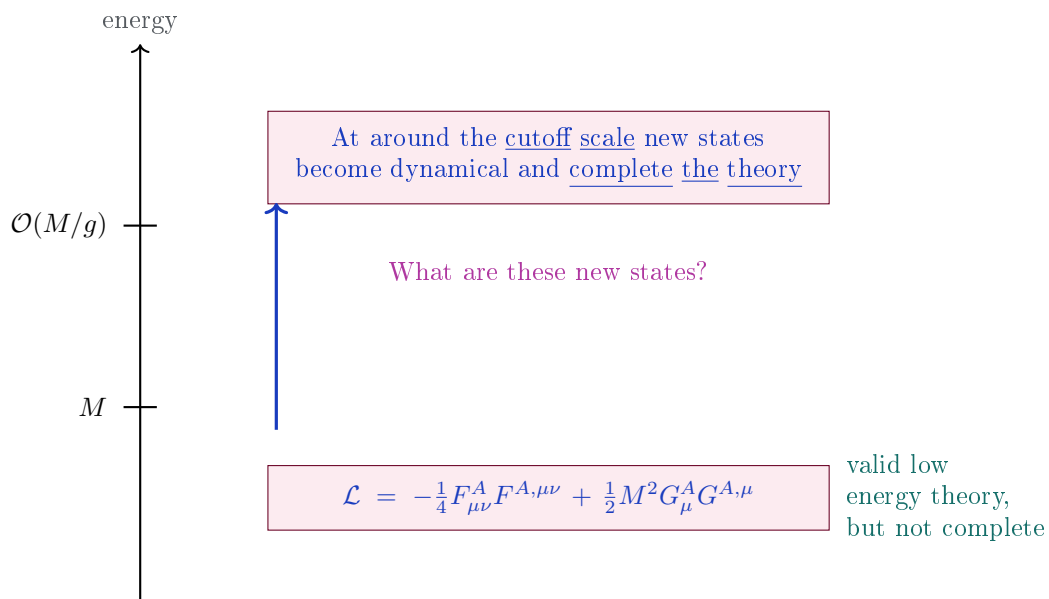
In this limit, $\varepsilon_L^\mu(p) \sim p^\mu/M$

A direct computation shows that, in this limit

$$i\mathcal{M} = \mathcal{O}\left(\frac{g^2 E^2}{M^2}\right)$$

$\triangle$ The amplitude grows with energy, and violates unitarity at energies $E = \mathcal{O}(M/g)$

$\rightsquigarrow$ We cannot just “put by hand” a mass term, because the theory we get is not complete



Where the growth comes from, and why it is fatal

The longitudinal polarization is the culprit. A massless vector has two transverse polarizations; a massive one has three. The extra, longitudinal mode has polarization vector

$$\varepsilon_L^\mu(p) = \left(\frac{|\vec{p}|}{M}, \frac{E}{M} \hat{p} \right) \xrightarrow{E \gg M} \frac{p^\mu}{M},$$

which *grows* with energy. Four of them in a $2 \rightarrow 2$ amplitude give $(E/M)^4$; the Yang–Mills vertices supply g^2 and, after the leading cancellations, the result is $g^2 E^2/M^2$. Setting this to $\mathcal{O}(1)$ gives the breakdown scale $E \sim M/g$.

Notice the pattern is *exactly* that of Lecture 21:

	amplitude	cutoff
Fermi theory	$G_F E^2$	$E \sim 1/\sqrt{G_F} \sim M_W/g$
massive Yang–Mills	$g^2 E^2/M^2$	$E \sim M/g$
gravity	E^2/M_P^2	$E \sim M_P$

In each case an amplitude growing like E^2 signals a missing degree of freedom, and the cutoff estimates where it lives.

Why massless Yang–Mills escapes this. There, gauge invariance forbids the longitudinal mode entirely — it is pure gauge, removable. The dangerous polarization simply is not in the spectrum, which is why Lecture 22 could conclude that the theory is renormalizable and complete. Give the boson a mass by hand and the mode becomes physical, with nothing organizing its interactions.

Why the abelian case survives. In massive QED the photon has a longitudinal mode too, but the photon does not self-interact — there are no g vertices to feed the growth. The diagrams above simply do not exist. This is the sharp sense in which the problem is specific to the *non-abelian* theory.

Where the scan stops — and what the answer is

The lecture ends on the question in magenta: *what are these new states?* It is left hanging deliberately, since the answer is the subject that naturally follows.

The resolution is not to abandon the mass but to obtain it *without* breaking gauge invariance — by breaking the symmetry *spontaneously*. One adds a scalar field with a potential whose minimum is not at the origin; the gauge symmetry of the Lagrangian is preserved, but the vacuum does not respect it. The would-be Goldstone bosons are then “eaten” to become the longitudinal polarizations of the gauge bosons, which acquire mass, while the surviving physical scalar — the Higgs boson — appears in the scattering diagrams above and cancels the dangerous E^2 growth.

Two things worth noting in advance. First, this is a genuine *prediction* of the argument on this page: something must show up at or below $\sim M/g$, and for $M \sim M_W \sim 80$ GeV with $g \sim 0.65$ that is a few hundred GeV. The Higgs boson was found at 125 GeV. Second, in the Standard Model the mechanism breaks $SU(2) \times U(1) \rightarrow U(1)_{\text{em}}$, giving mass to $W^\pm$ and Z while leaving the photon massless — which is precisely why the weak force is short-ranged and electromagnetism is not.

So the failure documented on [p. 36^{*****}] is not a dead end. It is the argument that forces the Higgs mechanism into existence, in the same way that the failure of linearized gravity in Lecture 19 forced general relativity.

Summary of Lecture 23 — the logic in nine lines

1. For a *simple* group the adjoint representation is irreducible; for a *semi-simple* group it is reducible, being the direct sum of the adjoints of the simple factors.
2. Concretely, $f_{ABC} = 0$ between different factors, so the transformation matrix acting on $F_{\mu\nu}^A$ is block diagonal, with one $\dim G_i \times \dim G_i$ block per factor.
3. Each block gives a separate invariant $\sum_A F_i^A F_i^A$, hence $\mathcal{L}_{YM} = \sum_i -\frac{1}{4g_i^2} F_i^A F_i^A$ with m *independent* couplings $g_1, \dots, g_m$ — nothing in the theory relates them.
4. Example: $SU(2) \times SU(3)$ has $\dim G = 11$, so 3 gauge bosons W_μ^A plus 8 gauge bosons G_μ^A , and two couplings g_2, g_3 .

5. Each additional $U(1)$ factor simply contributes one more gauge boson with the ordinary abelian field strength and its own coupling.
6. Matter charged under several factors carries one index per factor and lives in a tensor product $R_1 \otimes R_2$; the covariant derivative accumulates one term per factor, $D_\mu \Phi = \partial_\mu \Phi + ig_1 W^A T_{R_1}^A \Phi + ig_2 G^A T_{R_2}^A \Phi + igq A_\mu \Phi$.
7. Fermi's four-fermion theory of neutron decay is non-renormalizable ($[G_F] = [M]^{-2}$), so it must be the low-energy limit of a heavy exchange. Its γ -structure says the mediator is spin-1; the fact that it converts $n \rightarrow p$ and $e \rightarrow \nu$ says it must be a *non-abelian* gauge boson — and it must be massive.
8. Adding a mass by hand, $\frac{1}{2}M^2 G_\mu^A G^{A,\mu}$, breaks gauge invariance. That alone is survivable, but the now-physical longitudinal polarization $\varepsilon_L \sim p/M$ makes the $2 \rightarrow 2$ amplitude grow as $g^2 E^2/M^2$, violating unitarity at $E \sim M/g$.
9. So massive Yang–Mills is at best a valid low-energy theory, incomplete above its cutoff: new states must appear there to complete it. Identifying them is the next problem.